

A FAST APPROXIMATION ALGORITHM FOR THE MULTICOVERING PROBLEM

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The multicovering problem is: $\text{MIN } cx$ subject to $Ax \geq b$, $x \in \{0, 1\}^n$, where A is a matrix whose elements are all zero or one and b is a vector of positive integers. We present a fast heuristic for this important class of problems and analyze its worst-case performance: the ratio of the heuristic value to the optimum does not exceed the maximum row sum of the matrix A . The heuristic algorithm also provides a feasible dual solution vector that is useful in generating lower bounds on the value of the optimum.

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The multicovering problem requires finding a least total cost collection of sets such that in a given set of elements each element is covered a prescribed number of times. Formally, the problem may be formulated as the following integer programming problem:

$$\begin{aligned} \text{MSC} = \min \quad & \sum_{j \in J} c_j x_j, \\ \text{s.t.} \quad & \sum_{j \in J} a_{ij} x_j \geq b_i, \quad i \in I, \\ & x_j \in \{0, 1\}, \quad j \in J, \end{aligned}$$

where J is interpreted as the set of sets, each column j of the matrix $A = (a_{ij})$ is the 0–1 incidence vector of the j th set, I is the set of elements to be ‘covered’ and b_i the number of times that element i must be covered. Note that without loss of generality we can always assume the vector b to be positive and integral (if a non-positive component exists, eliminate that row; if a nonintegral component exists, then round it up to the nearest integer).

The multicovering problem is a generalization of the covering problem which is the special case with $b_i = 1$, $i = 1, \dots, m$. The set covering problem is NP-complete

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(see Garey and Johnson [3] for definitions); hence so is the multicovering problem. Problems such as the multicovering problem come up in communication or distribution problems where reliability is important and it is not sufficient to ‘cover’ each element only once. In order to increase reliability an element may be ‘covered’ several times, so when one covering facility fails some backup still exists.

The literature on the multicovering problem includes an integer programming approach applied to a problem of radio communications in West Germany by van Slyke [7] and an optimal integer programming algorithm given by Hall and Hochbaum [4]. Worst case analyses of heuristics for this problem include a greedy heuristic analyzed by Dobson [2] and a linear programming heuristic analyzed by Hochbaum [5, 6], with worst case bounds that are incomparable (the first is proportional to the logarithm of the maximum column sum, the latter, to the maximum row sum). For a variation of the problem in which x_j is an unbounded nonnegative integer Xiaoming and Van Slyke [8] propose a heuristic with a prescribed worst case bound. The restriction of bounded variables is not handled by that heuristic.

Our purpose in this paper is to present an algorithm that delivers an approximate solution to the multicovering problem with a worst case error bound that is at most the maximum row sum in the matrix A . This bound is similar to the one reported by Hochbaum [5, 6], but it is derived without using an algorithm to solve a linear program first. The running time of this algorithm is $O(\max\{n, m\} \cdot n)$ where $n = |J|$; hence it is fast and independent of the magnitude of the components of b or c . This approach is a generalization of the one taken by Bar Yehuda and Even [1] that derived an approximate solution for the set covering problem, with the same worst case properties as the linear programming derived solution (by Hochbaum [5, 6]), but without explicitly solving a linear program. A byproduct of the heuristic algorithm is a feasible dual solution vector that might be used in a dual ascent or other dual approaches used in optimal integer programming algorithms to provide bounds. In another study (Hall and Hochbaum [4]) we make use of such dual solutions with very encouraging empirical results.

The Algorithm

The input to the algorithm is the matrix A and the vectors b and c . The output is COVER – the indices of the sets selected and a vector $(y_i, i=1 \dots m; v_j, j=1 \dots n)$ that will later be proved to constitute a feasible dual solution.

Step 0. (Initialize) $v_j = 0, \quad j \in J,$
 $y_i = 0, \quad i \in I,$
 COVER = $\emptyset$.

Step 1. Let $i \in I$. Let $c_k = \min\{c_j \mid j \in J\text{-COVER and } a_{ij} = 1\}$. (k is the minimum cost column covering row i .) If no such minimum exists, stop – the problem is infeasible.

Step 2. Set $y_i \leftarrow y_i + c_k$. $\text{COVER} \leftarrow \text{COVER} \cup \{k\}$. For all $j \in J$ such that $a_{ij} = 1$ set $c_j \leftarrow c_j - c_k$.
If $c_j < 0$ then $v_j \leftarrow v_j - c_j$ and $c_j \leftarrow 0$.

Step 3. Set $b_i \leftarrow b_i - a_{ik}$, $i = 1, \dots, m$.
For all i' s.t. $b_{i'} = 0$, $I \leftarrow I - \{i'\}$.
If $I = \emptyset$ stop, else go to Step 1.

The algorithm repeats Step 1 at most n times, since if following n iterations the set I is not yet empty, then there is no feasible solution. This could occur for instance if the amount of required coverage exceeds the number of covering sets, n . At each iteration there are at most $O(\max\{n, m\})$ operations resulting in a total complexity of $O(\max\{n, m\} \cdot n)$.

The output of the algorithm is the set COVER of indices of the selected sets that multicover all elements, or a statement that the problem is infeasible. We shall now prove the worst case bound on the delivered solution and the feasibility of the dual vector derived.

The worst case bound

We now formulate the linear programming relaxation of the multi-covering problem and its dual:

$$\begin{aligned} \text{MSC}^* &= \min \sum_{j=1}^n c_j x_j, \\ \text{s.t. } \sum_{j=1}^n a_{ij} x_j &\geq b_i, \quad i = 1, \dots, m, \\ x_j &\leq 1, \quad j = 1, \dots, n, \\ x_j &\geq 0, \quad j = 1, \dots, n. \end{aligned}$$

MSC^* obviously bounds from below the optimal value to the multicovering problem. The dual to the linear programming relaxation above reads:

$$\begin{aligned} \max \sum_{i=1}^m b_i y_i - \sum_{j=1}^n v_j, \\ \text{s.t. } \sum_{i=1}^m a_{ij} y_i - v_j &\leq c_j, \quad j = 1, \dots, n, \\ y_i, v_j &\geq 0, \quad i = 1, \dots, m; j = 1, \dots, n. \end{aligned}$$

In the proofs of the facts that follow we shall use the notation $S_j = \{i \mid a_{ij} = 1\}$, i.e., S_j denotes the j th set.

Fact 1. For each $j \in \text{COVER}$,

$$c_j = \sum_{i \in S_j} y_i - v_j.$$

Proof. By construction $c_j + v_j = \sum_{i \in S_j} y_i$.

Fact 2.

$$\sum_{i \in S_j} y_i \leq c_j + v_j \quad \text{for all } j \in J.$$

Proof. This follows from Fact 1 and from Step 1 since the minimum cost column is always selected.

Fact 3. The output of the algorithm $(y, v) = (\{y_i\}_{i=1}^m, \{v_j\}_{j=1}^n)$ is a feasible solution to the dual problem.

Proof. First y_i, v_j are always nonnegative. This follows since y_i is equal to a cost c_j during one of the iterations and the c_j 's are always maintained as nonnegative numbers. Each v_j is a sum of positive numbers and hence nonnegative as well. Finally, Fact 2 establishes the feasibility with respect to the constraints.

Fact 4.

$$\sum_{i \in I} b_i y_i \leq \text{MSC}^* + \sum_{j \in J} v_j.$$

Proof. From Fact 3, (y, v) is a feasible dual solution, hence the weak duality theorem applies:

$$\sum_{i \in I} b_i y_i \leq \min \left(\sum_{j \in J} c_j x_j \right) + \sum_{j \in J} v_j \leq \text{MSC}^* + \sum_{j \in J} v_j.$$

Fact 5. $v_j = 0$ for all $j \in J\text{-COVER}$.

This fact follows from the selection made at Step 1 of the algorithm.
The following lemma is useful in the proof of the main theorem.

Lemma.

$$\sum_{j \in \text{COVER}} v_j \leq \sum_{i \in I} (b_i - 1) y_i \quad (\text{note that } b_i \geq 1, i \in I).$$

Proof. The value of the left-hand side and the right-hand side of the inequality vary during the algorithm's iterations.

We let the value of v_j and y_i after iteration t be denoted by $v_j^{(t)}$, $y_i^{(t)}$ respectively. Let T be the number of iterations. We shall prove by induction on t that

$$\sum_{j \in \text{COVER}} v_j^{(t)} \leq \sum_{i \in I} (b_i - 1) y_i^{(t)}, \quad t = 1, \dots, T.$$

For $t = 1$ the left-hand side is zero and the right-hand side nonnegative. We shall assume by induction that the inequality holds for $t = 1, \dots, l-1$ and prove for l .

Let $M = c_k$ be the minimum column cost selected at iteration l ; then the right-hand side increases by $(b_i - 1) \cdot M$ with y_i increasing by M . Each $v_j^{(l)}$ might be increased by at most M compared to the previous iteration but for no more than $b_i - 1$ columns. This is the case since a cost of a column could become negative (thus triggering the increase in $v_j^{(l)}$), only if it is already in the cover. If there were already b_i columns or more covering row i in COVER, then this row would have been removed from the set I , and thus could not be considered at iteration l . Therefore the inequality is preserved at each iteration and hence the desired result.

Theorem 1.

$$\sum_{j \in \text{COVER}} c_j \leq f \cdot \text{MSC}, \quad \text{where } f = \max_{i \in I} \left(\sum_{j \in J} a_{ij} \right).$$

Proof. From Fact 4 and the Lemma

$$\sum_{i \in I} y_i \leq \text{MSC}^* + \sum_{j \in J - \text{COVER}} v_j.$$

From Fact 5, the right-hand side of the above inequality is simply equal to MSC^* , hence

$$\sum_{i \in I} y_i \leq \text{MSC}^*. \quad (*)$$

Using Fact 1:

$$\begin{aligned} \sum_{j \in \text{COVER}} c_j + \sum_{j \in \text{COVER}} v_j &= \sum_{j \in \text{COVER}} \sum_{i \in S_j} y_i \\ &= \sum_{j \in \text{COVER}} \sum_{i \in I} a_{ij} y_i = \sum_{i \in I} \left(\sum_{j \in \text{COVER}} a_{ij} \right) y_i \\ &\leq \left(\max_{i \in I} \sum_{j \in \text{COVER}} a_{ij} \right) \cdot \sum_{i \in I} y_i \leq f \cdot \text{MSC}^*. \end{aligned}$$

The last inequality follows from (*) and the definition of f . Recalling that the v_j 's are nonnegative and that $\text{MSC}^* \leq \text{MSC}$ we derive the stated result. $\square$

From the proof of the theorem it follows that the heuristic solution value does not in fact exceed $(\max_{i \in I} \sum_{j \in \text{COVER}} a_{ij})$ times the value of the optimum. This factor could be arbitrarily smaller than $f = \max_{i \in I} \sum_{j \in J} a_{ij}$.

The derivation of the dual vector as a by-product of the heuristic also provides a certificate of optimality for the selected set COVER (or any other solution) satisfying

$$\sum_{j \in \text{COVER}} c_j = \sum_{i=1}^n b_i y_i - \sum_{j=1}^n v_j.$$

Even without certifying optimality, a dual solution provided valuable information to speed up the derivation of an integer programming optimal solution. Our work in that context is described in another paper (Hall and Hochbaum [4]).

Tightness of the bound

The worst case bound can be attained for the set covering problem which is a special case. Hence, it is obviously attainable for the multi-covering problem. An example that establishes this has the matrix A upper triangular, i.e., $a_{ij}=1$ for $i=1, \dots, j$ and $a_{ij}=0$ for $i>j$. Assume that $c=1_n$ and $b=1_m$ (vectors of n 1's and m 1's respectively). The optimal solution has $x_n=1$ with value 1. The heuristic solution value is $x=1_n$ and its value is n , hence the ratio is n which is also the maximum row sum. Note that the value of the dual solution in this case is 1. Furthermore, the linear programming heuristic for this problem reported in Hochbaum [6] will achieve the optimal solution in this case, though the worst case performance is identical to that of the algorithm reported here.

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